

Treasury Cash Flow Forecasting

AI-Powered Forecasting System for Aperam

1. Executive Summary

This project aims to develop an AI-powered cash flow forecasting system for Aperam's Treasury team. Aperam treasury manages €60M+ in weekly trade receivables and €50M+ in trade payables across European entities. Current manual forecasting from individual entities achieves ~90% Week 1 accuracy but declines to 64-83% by Week 4, with no visibility beyond 4 weeks and limits optimal liquidity decisions.

Europe Forecast Accuracy (%)				
Forecast Horizon	Receivables		Payables	
	Weighted AVG	ST Dév	Weighted AVG	ST Dév
Week 1	92%	6%	88%	27%
Week 2	86%	20%	89%	31%
Week 3	69%	18%	87%	26%
Week 4	64%	17%	83%	27%
Total	78%	15%	87%	28%

*AVG of 62M receivables and 50M payables

Tab 1. 2025 cash flow forecast accuracy metrics

The primary objective is to improve forecast accuracy from the current 70-90% baseline to 80-95% across an 8-week horizon, with the goal of reducing borrowing costs by €XXM annually through more accurate cash flow predictions. The system will use machine learning models (XGBoost, LightGBM, LSTM etc.) to predict weekly cash flows for Trade Receivables (TRR) and Trade Payables (TRP) Reval liquidity groups. The approach emphasizes enhancing existing treasury processes rather than replacing them, with rigorous backtesting and a phased pilot implementation.

Horizon	Current State	Target Accuracy	Value
Weeks 1-2	88-92%	90-95%	Modest improvement
Weeks 3-4	64-83%	80-90%	10-20% improvement
Weeks 5-8	No forecast	75-85%	New capability

Tab 2. Success Metrics

Forecast accuracy will be measured using WAPE (primary) and MAE/RMSE (secondary) at weekly horizons (W1–W8).

2. Problem Statement & Business Context

2.1 Current Challenges

Treasury manages complex weekly cash flows across Europe with decisions about credit lines, investments and funding depending upon the forecasts. The treasury team currently achieves high accuracy in Week 1 forecasts (approximately 90%) but experiences significant accuracy degradation over longer horizons, dropping to 74% by Week 4. This limits the treasury team's ability to:

- Optimize borrowing costs through better liquidity planning
- Make informed decisions about excess cash investment
- Arrange credit lines proactively rather than reactively
- Manage working capital more efficiently across multiple entities and currencies

2.2 Business Objectives

- Extend forecast horizon from 4 weeks to 8 weeks with reliable accuracy
- Reduce forecast error by 5-15 percentage points across all time horizons
- Decrease borrowing costs by €XXM annually through improved cash visibility
- Provide confidence intervals to quantify forecast uncertainty

2.3 Scope

Included	21 Aperam entities, Trade Receivables (TRR) + Trade Payables (TRP), 3 years daily data
Excluded	ELG & ASA divisions, other liquidity groups
Constraints	Weekly forecasts, no external market data in V1

Tab 3. Scope of the Project

3. Data Landscape

3.1 Primary Data Sources

The project leverages three primary data sources spanning Jan 2022 to Sep 2025.

Data Source	Description	Key Attributes
Actuals Data (Transactional)	Dataset with 625K daily transactions over 195 weeks	- Entity - Value Date - Amount - Liquidity Group
Liquidity Plan (Entity Forecasts)	4 week forward forecast for each entity	- Entity - Liquidity Group - Week - W1-W4 forecast
EuroFX Rates	Historical FX rates from European Central Bank	- Date - USD rate - CHF rate - Other currency pairs (EUR base)

Tab 4. Primary Data Sources

3.2 Historical Actuals Data Structure

Historical cash flow actuals are provided at the transaction level with the following structure:

Field Name	Data Type	Description	Example
Entity	String	Legal entity code	057, 17C7
Value Date	Date	Transaction date	2025-08-04
Amount Functional Currency	Decimal (EUR)	Transaction amount in EUR	€ 2,000,000
Liquidity Group	String	Cash flow category	TRR (receivables) TRP (payables)
Payment Currency	String	Original currency	EUR, USD, CHF

Tab 5. Actuals data structure

Note: Amount Functional Currency is already converted to EUR. The payment currency field indicates the original transaction currency for reference.

3.3 Liquidity Plan Data Structure

Liquidity plans represent the treasury's rolling 4-week forward forecasts. The data is provided in long format and transformed to wide format for modeling:

3.3.1 Raw LP Format:

Field	Description	Example
Entity	Entity Code	057
Year Title	Forecast reference week	2025/CW30
Liquidity Group	TRR or TRP	TRR
Item's Date	Forecast target week	08/04/2025
Amount in Plan Currency	Forecast value	100,000
Plan Currency	Currency of forecast	EUR, USD, CHF

Tab 6. Raw LP Format

3.3.2 Transformed LP Format:

Field	Description
Entity	Entity code
Year Title	Forecast reference week
Liquidity Group	TRR or TRP
W1_Forecast	Week 1 Forecast (EUR)
W2_Forecast	Week 2 Forecast (EUR)
W3_Forecast	Week 3 Forecast (EUR)
W4_Forecast	Week 4 Forecast (EUR)

Tab 7. Transformed LP Format

Note: For non-EUR plan currencies (USD, CHF), amounts are converted to EUR using historical FX rates from `eurofxref-hist.csv` matched by Item's Date.

3.4 Exploratory Data Analysis (EDA)

The exploratory data analysis focused on two main data sources: the actuals dataset (625,320 rows of transaction data from Jan 2022 to Oct 2025) and the liquidity plans dataset (31,666 rows of treasury forecasts). The 'actuals' data involved 18 entities after filtering for receivables and payables, with varying data completeness, with some entities like V756 (ASSS Italy) showing comprehensive historical coverage while others like 82J (ASSS USA) exhibit only 41 weeks of actuals data. An unexpected finding was 15,981 weekend transactions, confirmed as legitimate after consulting the treasury team.

3.4.1 Entity Analysis

Entity analysis shows that only 15 of 18 filtered entities have the required min threshold 52 weeks of historical data for machine learning models, which need a full seasonal cycle. Three entities (82J, 20B2, 25A4) lack sufficient data due to pipeline issues, inconsistent input, or recent acquisition. These three will be excluded from model training, using their submitted liquidity plans as forecasts instead. This approach ensures full entity coverage while allowing for future inclusion in model training as more data becomes available.

Category	Count	Data Coverage	Treatment
Tier 1: Full ML Modeling	15	52+ weeks	Ensemble Models (XGB, LGB, LSTM)
Tier 2: Pass-through	3 (82J, 20B2, 25A4)	< 52 weeks	Use treasury LP forecast; Reassess when more data is available

Tab 8. Entity Split

3.4.2 Liquidity Plan Characteristics

The liquidity plan dataset shows structural patterns that help with feature engineering and data preprocessing. Entities create rolling four-week forecasts, and these predictions reveal how forecast confidence changes over time. Not all entities provide a full four-week forecast, reflecting different forecasting practices. For missing forecasts, we will preserve as NaN as the tree based models would be able to handle those. Additionally, entities have specific patterns, with some having only Trade Receivables and others Trade Payables. The modeling approach needs to consider these entity-specific behaviors through "entity profiling" instead of treating all entities the same.

3.5 Data Quality Governance

The project focuses on continuous data quality monitoring, moving beyond one-time discoveries from initial analysis. This involves systematic validation processes before each model retraining cycle, including checking data continuity, correct currency conversion, and entity data coverage. The process requires treasury review of any new anomalies before implementing automated corrections, preventing the removal of legitimate business information.

Documentation of all data quality decisions and entity treatment rules is a critical component. This ensures clarity for future teams on decisions like pass-through forecasting for certain entities, retaining weekend transactions, and currency conversion logic, preventing regressions and revisiting settled decisions. The decision log in this document, along with code comments and operational runbooks, will guide ongoing model maintenance.

4. Solution Architecture

4.1 Core Philosophy

Our solution will enhance treasury expertise rather than replacing it. We integrate treasury liquidity plans as key inputs, leveraging human knowledge of upcoming events and operational changes alongside historical data patterns (seasonality, calendar effects, entity-specific behaviors). This combined approach should produce more accurate forecasts than either method alone. The system will be designed to be iterative, delivering value with current data while allowing for future enhancements as more data sources become available.

4.2 Feature Engineering

Our feature engineering will aim to balance complexity with data limitations (weekly aggregated transactions). We focus on temporal patterns, entity traits, calendar effects, and treasury forecasts (liquidity plans), ~ 100 features including lag variables, rolling statistics, calendar indicators, and trend measurements.

A key challenge is that simple weekly aggregation loses daily transaction pattern details (e.g., month-end spikes). To address this, we'll keep daily data during early feature engineering to extract sophisticated temporal features, then aggregate to weekly. These daily-to-weekly features capture insights like the percentage of transactions on month-end or daily amount variations, providing richer signals than simple weekly sums.

4.2.1 Core Feature Categories

Here are the core feature categories that will be used in the AI-powered cash flow forecasting system:

- **Lag Features:** These are crucial for time series forecasting, using past data to predict the future. We'll use 52 weekly lags (t-1 to t-52) to capture immediate, monthly, and annual patterns. This 52-week horizon dictates our minimum data requirement for entities.
- **Rolling Statistics:** These summarize recent history to understand trends and volatility. We'll use 4-week and 12-week rolling averages to smooth noise and provide context, and 26-week rolling standard deviations to measure volatility. These features complement lag features, offering the model flexibility.
- **Calendar Features:** These capture recurring temporal patterns like annual seasonality (week-of-year), monthly effects (payroll), and quarter-end impacts. These features apply broadly, independent of entity or historical patterns.
- **Trend Features:** These measure whether cash flows are increasing, decreasing, or stable. Linear trend slopes over 12 weeks quantify momentum, and trend acceleration metrics signal inflection points. These are valuable for operational changes, helping the model detect shifts in patterns.
- **Treasury liquidity plan forecasts** (Week 1-4) are incorporated as features. These capture expert expectations based on operational knowledge and events not in historical data. The model learns to weigh these projections based on their historical accuracy, trusting near-term forecasts more if they're reliable.
- **Entity Encoding:** Entity-specific patterns are encoded using one-hot encoding for the 15 Tier 1 entities. This allows the model to learn unique base levels and seasonalities for each entity. Gradient boosting models automatically learn feature interactions (e.g., how treasury forecasts affect specific entities or how month-end impacts receivables vs. payables) through their tree structure, without manual engineering. Category encoding (Trade Receivables vs. Trade Payables) enables a single model to learn patterns across both, improving efficiency given the modest dataset size.

Category	Count	Examples
Lag Features	52	t-1 through t-52 (one year lookback)
Rolling Statistics	~18	4/12/26/52-week rolling mean, median, std, min, max, trend, and volatility
Calendar Features	~12	Week of the year, month-end, quarter-end flags
Daily Patterns	~8	count of transactions in the week, daily std dev
Trend Features	~8	acceleration metrics, momentum indicators
Liquidity Plans	4	LP_W1, LP_W2
Entity Encoding	16	15 entity one-hot + Liquidity group binary

Tab 9. Model Feature Categories

4.3 Recursive Forecasting Strategy

The forecasting architecture uses a recursive prediction method, where a single model predicts one week ahead, and that prediction then informs the next week's forecast, continuing for the full 8-week horizon. This approach naturally will show declining accuracy over longer horizons due to error compounding, providing transparent insights into forecast uncertainty for better decision-making.

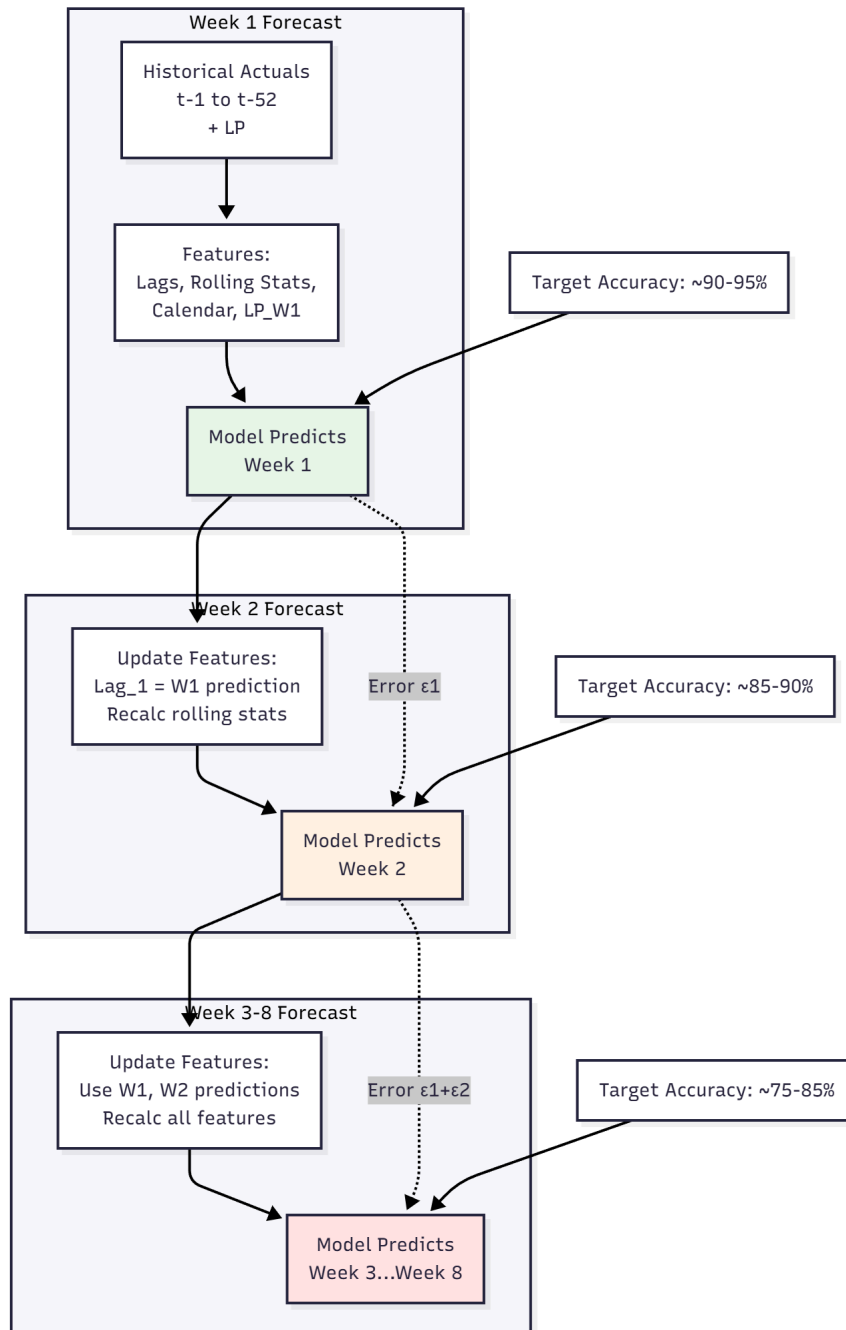


Fig 1. Recursive Forecasting Approach

4.4 Multi-Model Ensemble Architecture

The solution will employ an ensemble of three model types: XGBoost, LightGBM, and LSTM neural networks. Gradient boosted decision trees (XGBoost and LightGBM) are excellent at identifying complex patterns and interactions within features, handling missing data, and providing interpretable insights. LSTM neural networks complement these by modeling temporal dependencies in cash flow data. We will add a final forecast that's a simple average of the three models, which effectively prevents overfitting and provides robustness against individual model failures.

4.4.1 Stack Ensemble Design

Out-of-fold predictions from individual models (LightGBM, XGB...) will feed a ridge-regression meta-learner, producing horizon specific weighted outputs. Dynamic per entity weighting will adjust model contributions based on recent rolling performance to stabilize tail errors.

4.5 Production Architecture

The production system will operate on a weekly cadence, automatically extracting data from DENODO, retraining models, generating forecasts, and delivering outputs to treasury stakeholders.

4.5.1 Data Flow

- Source Layer: REVAL (core TMS) feeds daily transactions to SAS data warehouse
- Data Infrastructure: SAS performs ETL to DENODO virtual data layer
- ML Pipeline: Azure/AWS ML connects to DENODO weekly to extract fresh actuals and liquidity plans
- Processing: Automated data quality validation → feature engineering → model training/inference
- Output Delivery: Forecasts stored in central repository, could be exposed via Power BI dashboards and CSV exports

4.5.2 Key Components

Component	Purpose	Frequency	Critical Features
Data Extraction	Pull Actuals + LP from DENODO	Weekly (Tue noon)	Incremental load, retry logic
Quality Checks	Validate entities, gaps, currency etc	Per extraction	Auto-alerts on failures

Feature Engineering	Generate ~100 model features	Per training run	Preserve patterns
Model Training	Retrain models on latest data	Weekly/On-demand	Time series CV, hyperparameter tuning
Model Registry	Version control	Continuous	Automatic rollback if accuracy declines beyond a threshold
Inference Pipeline	Generate Forecasts	Weekly (post-training)	Recursive prediction, Confidence intervals
Monitoring	Track accuracy	Continuous	Eg: Alert if Week1 accuracy < 85%
BI Dashboard	Interactive forecast exploration	Real-time updates	Entity drill down, Accuracy trends
CSV/Raw File exports	Forecast output files	Weekly	

Tab 10. Key Components of the System

4.5.3 Monitoring & Alerts

- Data Quality Alert: Immediate notification if extraction fails or validation errors detected
- Model Performance Alert: Daily email if Week 1 forecast accuracy drops below 85% threshold
- Drift Alert: Weekly report flagging entities with degrading accuracy trends (>15% decline over 4 weeks)

4.5.4 Fallback Strategy

- If model training fails: Use previous week's model for inference
- If inference fails for specific entity: Use treasury's submitted LP as pass-through forecast
- If complete system failure: Treasury reverts to manual process with XX-hour resolution SLA

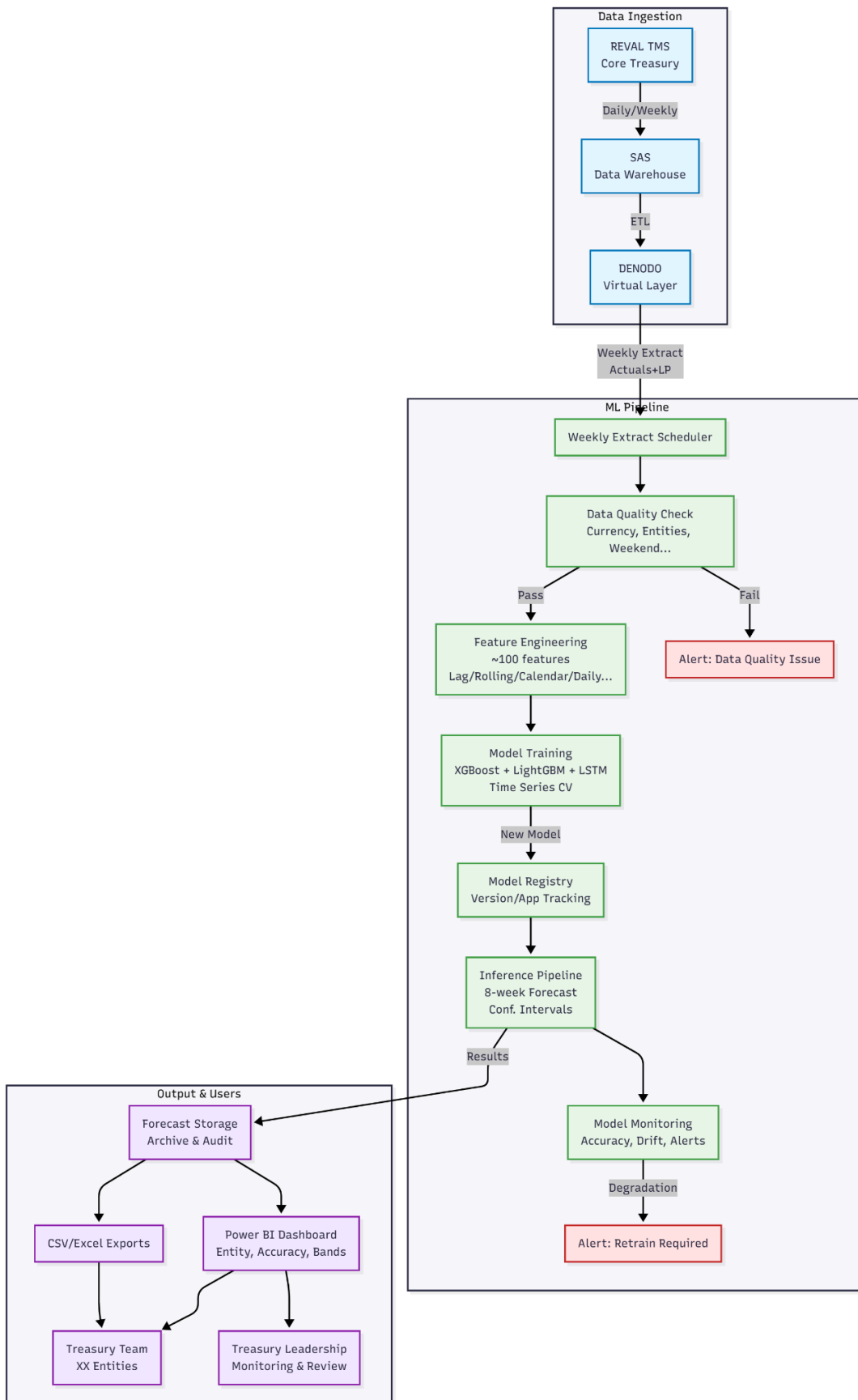


Fig 2. System Architecture

4.6. Alternate Horizon Modeling

To mitigate error compounding, a direct multi-horizon modeling approach will be evaluated in parallel. Comparative back testing will determine whether recursive or direct models yield superior horizon stability and performance.

4.7. Deployment Strategy

The deployment follows a 3-phase approach with strict go/no-go criteria at each gate, minimizing risk while enabling rapid feedback cycles.

Phase	Duration	Entities	Usage	Success Criteria
Shadow mode	XX weeks	All 15 Tier 1	View only	Forecasts generated successfully
Pilot	XX weeks	All 15 Tier 1	Secondary validation	Meet Accuracy targets
Production Rollout	-	All 15 Tier 1 + 3 Tier 2	Primary decision input	Meet Accuracy targets

Tab 11. Deployment Strategy

5. Success Metrics and Validation Framework

5.1 Accuracy Targets

Our project aims to significantly improve cash flow forecast accuracy over an extended 8-week horizon, surpassing current manual performance. Targets are set with the understanding that confidence naturally declines over longer periods.

- **Weeks 1-2:** We target 90-95% accuracy (WAPE), a modest but achievable improvement over the current 88-92% for Week 1, extending comparable quality to Week 2.
- **Weeks 3-4:** Targets range from 80-90% accuracy, a meaningful leap beyond the current 64-83% by Week 4.
- **Weeks 5-8:** We aim for ~75-85% accuracy, providing data-driven forecasts where none currently exist. Even at the lower end, this visibility should enable better long-term investment and facility planning.

5.2 Multi-Metric Evaluation

No single metric fully captures forecast quality, so we track five complementary metrics:

Metric	Description
WAPE	Primary metric. Weighted Absolute % Error - emphasizes accuracy on large flows, robust to near-zero values
MAE	Mean Absolute Error in EUR - shows typical error magnitude in business units
RMSE	Root Mean Squared Error - penalizes large outlier errors that cause business problems
Directional	% correct positive/negative predictions - critical for borrow vs invest decisions
MAPE	Mean Absolute % Error - reported for large values only (can be unreliable with near-zero denominators)

Tab 11. Model Evaluation Metrics

All metrics are disaggregated by forecast week (1-8), liquidity category (Trade Receivables vs. Trade Payables), and entity tier (full ML vs. pass-through) for granular performance insights.

5.3 Validation Approach

Our validation strategy combines technical rigor with practical business insights:

- **Time Series Cross-Validation:** We use an expanding window approach, training on all historical data up to a cutoff and testing on the subsequent XX weeks. This mimics production deployment and critically ensures recursive forecasting is properly implemented, using predictions rather than actuals for future weeks.
- **Business Validation and Pilot Testing:** This involves treasury workshops to review sample forecasts, assessing alignment with operational reality. A XX-week pilot will run model forecasts in parallel with manual liquidity plans, allowing the treasury team to build confidence, understand model behavior, and identify areas where the model might miss crucial context. Success is measured by the treasury team's willingness to use the forecasts for real decisions, supported by feature importance visualizations for explainability.

6. Decision Log

Critical decisions made during EDA and planning phases, documenting rationale for future reference.

Decision	Rationale
Exclude entities with <52 weeks actuals from ML training	Need full annual cycle for seasonal pattern learning. Entities 82J, 20B2, 25A4 use pass-through until sufficient history.
Retain weekend transactions (15,981 records)	Treasury confirmed legitimate business activity. Weekly aggregation handles naturally.
Preserve missing LP forecasts as NaN (don't impute)	Incomplete forecast arrays reflect actual behavior, not data errors. Tree models handle natively.
Fix currency conversion (use historical euroFX rates)	Non-EUR plans had FX=1.0. Daily historical rates ensure true EUR conversion.
Recursive 1-week forecasting	Mirrors production deployment, measures error compounding. Avoids train-test skew.
Unified model with category feature (not separate TRR/TRP models)	Pools information across categories, improves efficiency with modest dataset

Include treasury LP forecasts as model features	Leverages domain knowledge about upcoming events. Model learns appropriate weighting.
Entity-liquidity group profiling	Treasury revealed some entities have either TRR or TRP. Profile these behaviors. (refer Appendix A)

Tab 12. Decision Log

7. Appendix

Appendix A: Entity characteristics

Entity	Entity Name	Comments
057	ASSS LUXEMBOURG	-
11G5	RECYCO	This entity has usually just TRP
14C1	APERAM SOURCING SCA	This entity has usually just TRP
17C7	APERAM S.A.	This entity has usually just TRP
20B2	ASB Recycling	No LP inputs for this entity at the moment — coming soon.
25A4	UNIVERSAL STAINLESS & ALLOY PRODUCTS INC. - CORPORATE	Recently acquired entity. They start to fill LP recently
4B9	ASSS TUBES CZ	Sometimes this entity fails to provide LP updates.
82J	ASSS USA	In the past, they were not consistent in providing weekly LP updates. However, they are now reporting LP every week and are back on track..

86W	APERAM STAINLESS EUROPE	This entity usually has TRR, with only a very small amount of payables.
97C	APERAM ALLOYS IMPHY	-
97L	ASSS SWITZERLAND	-
97R	ASSS POLAND	-
98Y	ASSS CANADA	-
T055	APERAM STAINLESS BELGIUM DIS	This entity usually has TRR, with only a very small amount of payables.
T056	APERAM STAINLESS BELGIUM IND	This entity has usually just TRP
V002	APERAM STAINLESS FRANCE	This entity has usually just TRP
V265	ASSS PRECISION	This entity usually has TRR, with only a very small amount of payables.
V508	ASSS FRANCE	-
V751	ASSS IBERICA	-
V756	ASSS ITALY	-
V798	ASSS GERMANY	-

Appendix B: Liquidity Plans - Entity Coverage

Entity	First Date	Last Date	Total_Rows	Unique_Weeks	Data_Span_Days	Data_Span_Years	Has_Limited_Data
25A4	10/03/2025	17/11/2025	232	29	252	0.689938	TRUE
20B2	01/07/2024	12/05/2025	11	5	315	0.862423	TRUE
4B9	03/01/2022	10/11/2025	1417	197	1407	3.852156	FALSE
57	03/01/2022	17/11/2025	1570	202	1414	3.871321	FALSE
17C7	03/01/2022	17/11/2025	801	202	1414	3.871321	FALSE
14C1	03/01/2022	17/11/2025	794	202	1414	3.871321	FALSE
11G5	03/01/2022	17/11/2025	796	202	1414	3.871321	FALSE
82J	03/01/2022	17/11/2025	1082	147	1414	3.871321	FALSE
86W	03/01/2022	17/11/2025	1084	202	1414	3.871321	FALSE
97C	03/01/2022	17/11/2025	1588	202	1414	3.871321	FALSE
97L	03/01/2022	17/11/2025	1525	202	1414	3.871321	FALSE
97R	03/01/2022	17/11/2025	1587	202	1414	3.871321	FALSE
98Y	03/01/2022	17/11/2025	1510	197	1414	3.871321	FALSE
T055	03/01/2022	17/11/2025	1460	202	1414	3.871321	FALSE
T056	03/01/2022	17/11/2025	799	202	1414	3.871321	FALSE
V002	03/01/2022	17/11/2025	806	202	1414	3.871321	FALSE
V265	03/01/2022	17/11/2025	1586	202	1414	3.871321	FALSE
V508	03/01/2022	17/11/2025	1411	202	1414	3.871321	FALSE
V751	03/01/2022	17/11/2025	1576	202	1414	3.871321	FALSE
V756	03/01/2022	17/11/2025	1586	202	1414	3.871321	FALSE
V798	03/01/2022	17/11/2025	1589	202	1414	3.871321	FALSE

Appendix C: Liquidity Plans Data Structure

```
[12]: lp_df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 31666 entries, 0 to 31665
Data columns (total 50 columns):
#   Column                                                                 Non-Null Count  Dtype
---  -
0   PLAN-ID                                                                31666 non-null  int64
1   Item-ID                                                                31666 non-null  object
2   Entity                                                                31666 non-null  object
3   Group Description                                                      31666 non-null  object
4   Liquidity Group/Super Liquidity Group                                31666 non-null  object
5   Item's Date                                                            31666 non-null  object
6   Amount                                                                31666 non-null  float64
7   Currency                                                              31666 non-null  object
8   Credit/Debit                                                          31666 non-null  object
9   Origin                                                                31666 non-null  object
10  Item Status                                                            31666 non-null  object
11  Plan Name                                                             31666 non-null  object
12  Year Title                                                            31666 non-null  object
13  Plan Currency                                                         31666 non-null  object
14  Plan Status                                                            31666 non-null  object
15  Rate                                                                  31666 non-null  float64
16  Amount in plan currency                                              31666 non-null  float64
17  Project                                                                0 non-null      float64
18  Contra Entity                                                         0 non-null      float64
19  Cash Transaction Reason                                              1541 non-null   object
20  Cash Transaction Type                                                0 non-null      float64
21  From Actual Data                                                     31666 non-null  object
22  Rate Type                                                            31666 non-null  object
23  Rate Type Description                                                31666 non-null  object
24  Record Number of Cash Transaction                                    0 non-null      float64
25  Cash Flow Number                                                     31666 non-null  int64
26  Actual Data Internal Reference 1                                     0 non-null      float64
27  Actual Data Internal Reference 2                                     0 non-null      float64
28  Actual Data Cash Flow Ref.                                           0 non-null      float64
29  Counterpart                                                           0 non-null      float64
30  PTP Account                                                           0 non-null      float64
31  Account                                                              0 non-null      float64
32  Long Name of Account                                                 0 non-null      float64
33  Comment                                                              1795 non-null   object
34  Created by                                                            31666 non-null  object
35  Created at                                                            31666 non-null  object
36  Changed by                                                            675 non-null    object
37  Changed at                                                            675 non-null    object
38  Plan Category                                                         31666 non-null  object
39  Plan Type                                                             31666 non-null  object
40  Consolidation possible                                               31666 non-null  object
41  Exposure Aggregation                                                 31666 non-null  object
42  Entity ID                                                            31666 non-null  object
43  Entity Name                                                           31666 non-null  object
44  Entity Short Name                                                    31666 non-null  object
45  Group ID                                                             31666 non-null  object
46  Type                                                                  31666 non-null  object
47  Project ID                                                            0 non-null      float64
48  Super Liquidity Group                                                31666 non-null  object
49  Original Liquidity Group                                             0 non-null      float64
dtypes: float64(16), int64(2), object(32)
memory usage: 12.1+ MB
```

Appendix D: Liquidity Plans - EDA

